Week 12: 16th – 20th November 2026

Subject: Mathematics Diploma Programme **Unit Title: Functions, Models and Complex Numbers** **Grade: 11**

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Differentiation	Resources and links for Parents
Lesson 1: Quadratic Functions and Key Features	I can analyse quadratic functions by identifying and interpreting their key features from equations and graphs.	2.1 Clarify the meaning of vertex, axis of symmetry, intercepts, zeros/roots and maximum/minimum values using annotated graphs and colour-coded examples.	**Related Videos / Resources** Drawing and sketching graphs [IB Maths AI SL/HL] Solving using technology [IB Maths AI SL/HL] Modelling in general [IB Maths AI SL/HL] Quadratic models intro [IB Maths AI SL/HL] Quadratic models example [IB Maths AI SL/HL]
Lesson 2: Quadratic Models and Applications	I can construct, use and interpret quadratic models to represent real-world situations and solve problems involving roots and maximum or minimum values.	5.3 Provide a structured modelling scaffold: define variables, form/select a model, solve and interpret; gradually remove prompts.	
Lesson 3 (SL): Grow Day – Reflection	I can reflect on my learning, identify strengths and gaps in my understanding, and use feedback and practice to improve my mathematical skills.	Use selected practice questions to identify areas of confidence and topics requiring further support. Analyse errors in previous work and correct solutions, explaining the mathematical reasoning. Set an individual improvement target and complete a focused practice task. Reflection: identify one strength, one area for improvement and a specific next step.	
Lesson 4 (HL): Complex Numbers – Powers and Roots in Euler Form	I can represent complex numbers in polar and Euler forms and use these forms to calculate powers and roots by applying De Moivre's theorem.	8.2 Extend confident learners to multi-step HL problems involving all roots and their geometric representation on an Argand diagram.	
Homework	Quadratic functions and models as well as complex numbers assigned on InThinking.		
Teacher Notes	Students should bring a GDC/graphing calculator, notebook and device. Ensure access to lesson PowerPoints, Teams/OneNote resources and practice materials.		

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Technology Integration	Microsoft Teams; OneNote; GDC/graphing calculator; Desmos; lesson PowerPoints; digital practice materials. Use graphing technology to analyse quadratic functions and models, spreadsheets/GDC for exponential growth, and a complex-number calculator to verify polar/Euler forms, powers and roots.

Cross-curricular links	Physics: quadratic models for projectile motion and parabolic paths. Economics/Business: revenue, profit and optimisation using quadratic models. Grow Day: reflect on mathematical learning strategies, feedback and progress. Physics/Engineering: complex numbers in electrical systems, waves and signal representation.
External Assessment Integration	IB Mathematics AI/DP: quadratic functions, key graph features, intersections and quadratic models. Grow Day: review and consolidate previously taught mathematical skills through reflection, error analysis and targeted practice. Mathematics HL: complex numbers in Cartesian, polar and Euler forms; De Moivre's theorem; powers and roots. Practise IB-style questions requiring accurate notation, technology use and contextual interpretation.
Wellbeing/ ATL Skills	Thinking Skills: analyse graph features, select suitable models, apply De Moivre's theorem and justify mathematical processes. Communication Skills: use correct notation, labelled graphs and clear contextual interpretations. Self-Management Skills: organise multi-step solutions, use technology appropriately and check the reasonableness of results. Reflection: evaluate model assumptions and limitations.
Standards	IB Mathematics AI/DP syllabus alignment: uadratic functions and key graph features; intersections using technology; quadratic models. Grow Day supports consolidation, self-assessment and reflection on previously taught content. Mathematics HL: complex numbers, modulus and argument, polar and Euler forms, powers and roots using De Moivre's theorem.
Inquiry/ TOK Question	How does the choice of mathematical representation help us understand and predict real-world behaviour? When are quadratic and exponential models useful, and what assumptions limit their accuracy? How does expressing a complex number in Euler form make powers and roots easier to determine?
Success Criteria (Measurable)	**Lesson 1: Quadratic Functions and Key Features** · All students must identify the vertex, axis of symmetry, intercepts and zeros/roots from a quadratic graph. · Most students should determine key features from a quadratic equation and explain the maximum or minimum value. · Some students could analyse the effects of changing parameters and justify their conclusions. **Lesson 2: Quadratic Models and Applications** · All students must identify variables and interpret a given quadratic model. · Most students should construct or select a suitable quadratic model and use it to solve a contextual problem. · Some students could evaluate the assumptions and limitations of a model and justify its suitability. **Lesson 3 (SL): Grow Day – Reflection** · All students must identify topics they understand well and areas where they need further practice. · Most students should analyse errors in their work, correct them and explain how to improve. · Some students could set a specific improvement target, select an effective strategy and demonstrate progress through independent practice. **Lesson 4 (HL): Complex Numbers – Powers and Roots in Euler Form** · All students must identify the modulus and argument and convert a complex number into polar and Euler forms. · Most students should use Euler form and De Moivre's theorem to calculate powers of complex numbers. · Some students could determine all roots of a complex number and represent them accurately on an Argand diagram.
National Identity Domain	Innovation
	Area: Citizenship
	Implementation: Connect quadratic models to UAE architecture, sports trajectories and business contexts. Use Grow Day to encourage responsibility for learning, reflection, perseverance and purposeful use of feedback. Highlight applications of complex numbers in engineering, electrical systems and telecommunications, reinforcing innovation and future-ready skills.